Multitemporal Intrinsic Image Decomposition With Temporal–Spatial Energy Constraints for Remote Sensing Image Analysis

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***Abstract*— Due to interference with remote imaging by some natural factors, the multitemporal analysis ability is limited by the spectral drift between images. In this article, a new approach to optimize the existing multitemporal analysis system is proposed: multitemporal intrinsic image decomposition (MIID). The MIID method is designed to extract common spectral reflectance from multitemporal images. With MIID, multitemporal classification, changing detection, and index extracting will become extremely easy and more accurate. Firstly, without considering land cover change, the general MIID framework is proposed by adding local temporal–spatial energy constraints in traditional intrinsic images decomposition. On this basis, an improved MIID method with change detection (CD) (CD-MIID) capability is proposed to make the model adapt to the land cover change situation. Finally, specific steps of how to use MIID methods in the multitemporal analysis are given. Multitemporal multispectral/hyperspectral remote sensing images from GF-1, GF-2, GF-5, Landsat TM, and two groups of captured datasets with reflectance truth map are used to evaluate the performance. The experimental results show the following two points: first, the MIID methods achieve better extraction results of spectral reflectance. Second, the proposed MIID methods have better performance both on multitemporal classification and CD.**

***Index Terms*— Change detection (CD), classification, intrinsic image decomposition, multitemporal remote sensing.**

I. Introduction

MULTITEMPORAL remote sensing image can record the change process of ground objects in detail; as a result, multitemporal data analysis can be widely used to monitor agriculture [1], [2], urban heat island [3], forest change [4], land cover [5], and missing data recovery [6]. As we know, the basis of the joint analysis of multitemporal remote sensing images is the consistency of data distribution. However, spectral signatures are subject to change over time due to some natural factors, like solar elevation (date, time), atmospheric, environment, relief (shadow and slope), phenology, and disease. This spectral change in different temporal images (change over time) is called spectral drift. Due to spectral drift, data distribution of the same ground cover object in different times is not consistent. This problem seriously degrades the performance of multitemporal analysis, including multitemporal classification and change detection (CD). In the same way, because of the natural factors mentioned above, the reflectance information extracted from each temporal image always has deviated, which also leads to the inaccuracy of quantitative index extraction.

In recent years, some alignment methods [including domain adaptation (DA) and transfer learning (TL)] have been used to overcome the inconsistent distribution between multitemporal remote sensing images. These alignment methods identify the direct relations between elements from different datasets firstly, and then they map different datasets into a common subspace to ensure that the mapped data have the same data distribution before using. Manifold alignment [7]–[11], kernel alignment [12], [13] and subspace alignment [14], [15] have been widely used to align different remote sensing datasets. However, there are some disadvantages to these alignment methods in multitemporal analysis. The most important aspect is that those methods cannot completely align multitemporal images by calculating the direct relation. There are still large differences in data distribution in the aligned images, when the difference between original multitemporal images is large.

Besides DA methods, relative radiometric normalization is also designed to solve the drift problem between multitemporal images. Relative radiometric normalization is mainly to reduce the radiometric influences caused by nonsurface factors and to ensure that the gray value differences between temporal images reflect actual changes on the surface of the earth [16]. Absolute and relative radiometric normalization methods are combined to simultaneously obtain precise surface reflectance values and reduce data shift between images [17]. Besides, polynomial regression with Gaussian weighted regression is also adopted to improve relative radiometric normalization [18].

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Deep learning has been successfully applied on DA, including cross image classification and CD. Using shallow network to initialize a new task network is equivalent to migrate existing knowledge into a new task, which is the simplest application of deep learning in knowledge transfer [19]. According to the different roles played by deep neural networks in DA methods, the deep DA methods can be divided into four categories: methods based on differences in domain distribution, adversarial-based methods, Reconstruction based methods, and sample generation-based methods [20]. The method based on the difference in domain distribution mainly extracts domain invariance features by minimizing the difference in the distribution of two domains, and the typical method is deep adaptation network (DAN) [21]; adversarial-based methods align distributions through adversarial learning, and the typical method is multiadversarial DA (MADA) [22]. Yang *et al*. [23] proposed a cross-scene CD framework by introducing adversarial learning to align the distributions of the selected source samples and the target ones; reconstruction-based methods achieve DA by suppressing information loss or decoupling features, and the typical method is domain separation network (DSN) [24]. Saha *et al*. [25] proposed a segmentation algorithm of multitemporal images based on a loss function which detects object segments from individual images and establishes a correspondence between distinct multitemporal segments. Generative adversarial network (GAN) based on sample generation can synthesize fake samples that conform to the target domain distribution to support training. The typical method is CycleGAN [26]. Saha *et al*. [27] proposed a multiple-CD method in VHR multisensor images by exploiting a CycleGAN that consists of two generator CNN networks to learn in an unsupervised way.

Assuming that we can eliminate the influence of natural factors and extract accurate and consistent spectral reflectance from multitemporal images (only the parts which the land cover does not change), it will effectively improve the current multitemporal analysis ability and become a new multitemporal analysis approach: firstly, it can effectively improve the accuracy of reflectance extraction and improve quantitative analysis. Secondly, if we can extract common reflectance from multitemporal images, then the common component is strictly aligned. At this time, multitemporal classification (using one image to classify another) is equivalent to the self-classification and will achieve a very high classification accuracy. The third is to improve the effect of CD. When there is obvious difference between the material before and after the change, and the spectral band is enough, it is easy to detect the change area by spectral variation. In other words, it is easier to capture the change of land cover only from the analysis of reflectance components.

Intrinsic image decomposition (IID) is designed to extract the reflectance and shading (or illumination) information from the image [28]. Since IID extracts two or more variables from one data at the same time, it is obvious that IID is a typical ill-posed inversion problem, and it is difficult to obtain effective extraction results. In order to improve the inversion accuracy, researchers mainly establish an effective IID model by adding prior hypothesis information or providing additional data. In terms of the establishment of prior hypotheses, some researchers think that the shading or light component is smooth in the local adjacent area, and thus builds the relevant decomposition model [28]. Some researchers think that the reflectance of objects in the local neighborhood is usually the same, and the reflectance of objects with the same type in the global is basically the same. Therefore, from the perspective of the whole image, the reflectance is sparse. Based on this assumption, some new IID methods are established in sparsity reflectance [29], [30]. Some researchers build decomposition models by using the above two assumptions at the same time [31]. In addition, manually labeled supervision information is also used to improve the inversion accuracy of the IID model [32]. In terms of adding additional data methods, some researchers add depth information (i.e., using RGB-D dataset) to simplify the intrinsic decomposition model [33]–[36], and adding multiple images under different illuminations [37], [38] (but mainly laboratory conditions). Intrinsic video decomposition [39], [40] also uses the extra information of multiple frames for decomposition. Recently, IID is extended and applied to remote sensing image decomposition and extract spectral reflectance, especially on hyperspectral image [41]–[43].

Although the IID methods can extract spectral reflectance know, IID is a typical ill-posed problem. Extracting spectral reflectance from a single remote sensing image cannot guarantee the accuracy of the extraction results in principle. Secondly, the complex imaging factors of remote sensing image will interfere with the decomposition model and reduce its performance. In order to improve the decomposition accuracy, it is necessary to increase the prior information input. Therefore, when the common spectral reflectance component is extracted from multitemporal images, more images become the prior information of model input. Therefore, the accuracy of the extracted reflectance component is higher than that of a single remote sensing image in principle.

Thus, building a multitemporal IID model to extract common spectral reflectance from multitemporal images can not only improve the accuracy of reflectance extraction but also provide a new analytical approach to improve the ability of multitemporal analysis (as mentioned above), including multitemporal classification and CD. In view of this, we extend the traditional IID framework for multitemporal cases. Firstly, a general multitemporal IID [without considering land cover change, multitemporal intrinsic image decomposition (MIID)] framework is proposed by adding local temporal–spatial energy constraints. On this basis, an improved MIID method with CD (CD-MIID) capability is proposed to make the model adapt to the land cover change situation. Finally, we give the specific steps of the CD-MIID model in the multitemporal analysis.

The remaining parts of this article are organized as follows. Section II describes the MIID method, including model hypothesis, decomposition-based local temporal–spatial energy constraints, and optimization solution. Section III describes a CD-MIID method under the change condition hypothesis and its application in multitemporal analysis. Section IV presents multitemporal datasets and experimental

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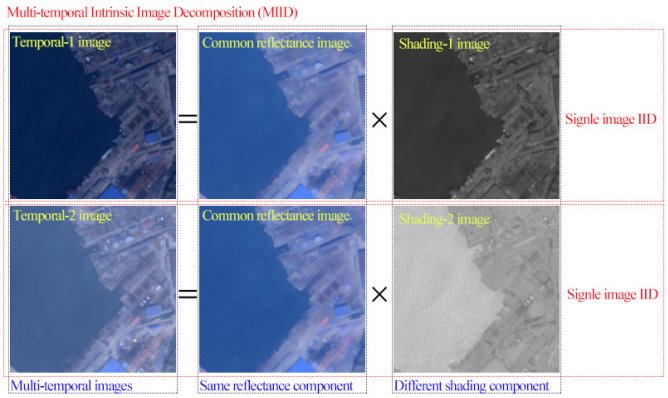


Fig. 1. MIID framework.

design. Section V summarizes experimental results in detail. Section VI draws conclusions for this article.

II. Multitemporal Intrinsic Image Decomposition

MIID is designed to overcome the spectral drift between images and extract the common reflectance image of invariant objects (without land cover changing) from two multitemporal images at different observation times (shown in Fig. 1). There are many advantages in realizing the MIID method. First, when more temporal images are input into the IID model, it provides more prior information, and more prior information ensures the improvement of spectral reflectance extraction accuracy in principle. Second, extracting a completely consistent spectral reflectance is equivalent to completely eliminate the spectral differences between images, so multitemporal classification (using source temporal image to classify the target temporal image) here is equivalent to the self-classification (i.e., internal classification) on the common spectral reflectance image.

So, we firstly proposed a basic MIID framework in this section which assumes that there is no change in land cover between the two images. This framework consists of two parts: hypothesis of the multitemporal image representation model, and decomposition constraints based on local temporal–spatial energy constraints.

A. MIID Hypothesis

For a single remote sensing image, the relationship between the components of IID is as follows:

$$I = sR \tag{1}$$

where I is the input image, R is the reflectance component, and s is the shading component (or illumination component).

For multitemporal remote sensing images, when there is only spectral drift between different images without land cover changes, the type of objects represented by each point in the image is unchanged, so the reflectance component R of the multitemporal images should be the same. Based on this, the intrinsic decomposition model of multitemporal images can be assumed as follows:

$$I_1 = s_1 R, \quad I_2 = s_2 R \tag{2}$$

where I_1 and I_2 represent the two input multitemporal images, R is the common reflectance component of the two images (I_1 and I_2), s_1 denotes the illumination component of I_1, and s_2 denotes the illumination component of I_2.

B. Decomposition-Based Local Temporal–Spatial Energy

For remote sensing images, the illumination and shadow conditions are basically the same in the local neighborhood, so the image intensity change in the local area mainly comes from its color difference. In other words, the pixels with the similar intensity values always share the similar reflectance values. Therefore, the reflectance value of one pixel can be represented well by the weighted summation of its neighborhood pixel values [32], [36]. In addition, for each temporal image, the error between the product of common reflectance and their own illumination and the original image should be as small as possible. Based on the concepts mentioned above, we can define a temporal–spatial energy function to obtain the intrinsic images (especially the intrinsic reflectance image) from the multitemporal input image. Here, we only assume that the input is two temporal images. The temporal–spatial energy function of MIID is written with three unknown variables (S_1, S_2, and R) and be expressed as follows:

$$\begin{aligned} E\left(R, \tilde{S}_1, \tilde{S}_2\right) = & \sum_{i \in P} \left(R_i - \sum_{j \in N(i)} w_{ij} R_i \right)^2 \\ & + \sum_{i \in P} \left((I_1)_i \left(\tilde{S}_1\right)_i - R_i \right)^2 \\ & + \sum_{i \in P} \left((I_2)_i \left(\tilde{S}_2\right)_i - R_i \right)^2 \end{aligned} \tag{3}$$

where P is the pixel number of input image (temporal-1 image or temporal-2 image), and $N(i)$ stands for the all pixels from local neighborhood of pixel i. Here, we use the circumscribed rectangle centered on the current pixel i to represent its corresponding neighborhood. The size of the circumscribed rectangle (i.e., width sigma) can be set as 3×3, 5×5, 7×7, and so on. We use 5×5 as the width sigma in all of the later experiments. The first term of (3) is the similarity constraint of reflectance in a local neighborhood. The basis is that the reflectance image is smooth and can be expressed linearly with adjacent points. The second term of (3) is the relationship constraints of (1) for temporal-1 image ($\tilde{S}_1 = 1/S_1$), and the third term of (3) is the relationship constraints of (1) for temporal-2 image ($\tilde{S}_1 = 1/S_1$). w_{ij} is the similarity of the reflectance value between pixel i and pixel j. w_{ij} is calculated by using the following formula:

$$w_{ij} = e^{-0.5\left(\frac{\sum_{b=1}^{B}\left(I_{ib} - I_{jb}\right)}{B\sigma}\right)^2} \tag{4}$$

where B is the number of spectral bands, b is the index of the current spectral band, and σ is the similarity control coefficient. Generally, the range of sigma is set between 0.5 and 10. In this article, σ is set as 1 in all experiments. The error of spectral value is used to calculate the similarity between two points. By optimizing (3), we can obtain the common intrinsic reflectance of the multitemporal images by

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using the following:

$$\arg\min_{R,\tilde{S}_1,\tilde{S}_2} E\left(R,\tilde{S}_1,\tilde{S}_2\right) \quad \forall i \in P \ \forall b \in [1,\ldots,B],\ 0 \le R_b \le 1. \tag{5}$$

The energy optimization of (5) is a typical quadratic equation based on the variables R_b, $\tilde{S}_1$, and $\tilde{S}_1$. By obtaining the first-order differentiation of each variable in (3), we can get

$$\frac{\partial E\left(R,\tilde{S}_1,\tilde{S}_2\right)}{\partial \tilde{S}_1} = 2\sum_{b=1}^{B} I_{1b}\left(\tilde{S}_1 I_{1b} - R_{1b}\right) \tag{6}$$

$$\frac{\partial E\left(R,\tilde{S}_1,\tilde{S}_2\right)}{\partial \tilde{S}_2} = 2\sum_{b=1}^{B} I_{2b}\left(\tilde{S}_2 I_{2b} - R_{2b}\right) \tag{7}$$

$$\begin{aligned}\frac{\partial E\left(R,\tilde{S}_1,\tilde{S}_2\right)}{\partial R_b} &= 2R_{ib}\left(3 - R_{ib}\sum_{j\in N(i)} w_{ij}R_{jb}\right) \\ &\quad + \sum_{j\in N(i)} w_{ji}\sum_{k\in N(j)} w_{jk}R_{kb} \\ &\quad - 2\tilde{S}_{1i}\cdot(I_{1b})_i - 2\tilde{S}_{2i}\cdot(I_{2b})_i \\ &\quad - 4R_{jb}\sum_{j\in N(i)} w_{ji} \quad \forall i \in P \\ &\quad \forall b \in [1,\ldots,B]\end{aligned} \tag{8}$$

$$\begin{aligned}\frac{\partial E\left(R,\tilde{S}_1,\tilde{S}_2\right)}{\partial \tilde{S}_1} &= \frac{\partial E\left(R,\tilde{S}_1,\tilde{S}_2\right)}{\partial \tilde{S}_2} = \frac{\partial E\left(R,\tilde{S}_1,\tilde{S}_2\right)}{\partial R_1} \\ &= \cdots = \frac{\partial E\left(R,\tilde{S}_1,\tilde{S}_2\right)}{\partial R_b} = \cdots \\ &= \frac{\partial E\left(R,\tilde{S}_1,\tilde{S}_2\right)}{\partial R_B} = 0.\end{aligned} \tag{9}$$

By finding the zero position of the first derivative of (6)–(8) [i.e., building (9)] and using Gaussian–Seidel method [44], we can optimize (3) and get the optimal solutions of MIID iteratively. The iteration functions of each unknown variable (R_b, $\tilde{S}_1$, and $\tilde{S}_1$) are shown as (10)–(12), shown at the bottom of the page.

In order to get faster convergence to the optimal solution, the initial values of parameters (R, S_1, S_2, and ε) can be set as follows: ε is set as 10^{-5}; S_1 is the average of each band image of temporal-1 image; S_2 is the average of each band image of temporal-2 image; and $R = (I_1/S_1 + I_2/S_2)/2$. In order to ensure calculation time is not too long, we usually set the maximum number of iterations, which is often set to 100.

It should be noted that the proposed MIID method cannot effectively deal with the shadow problem due to the cloud, the high spatial resolution, and high buildings. Partial shadow will cause the reflectance extraction in shadow area to be dark, which cannot be consistent with the reflectivity brightness of other nonshadow areas.

The detailed procedure of MIID is shown as follows.

Algorithm 1 Procedure of MIID

Input: Temporal-1 image I_1, Temporal-2 image I_2, $\varepsilon > 0$, P is the pixel number of input image

i) For each pixel $i \in P$, finding pixel i local neighborhood $N(i)$, calculate w_{ij} using (4).

ii) Repeat:

Calculate R_i by using (12)

Calculate $\tilde{S}_{1i}$ by using (10)

Calculate $\tilde{S}_{2i}$ by using (11)

Calculate E by using (3) with R_i, $\tilde{S}_{1i}$ and $\tilde{S}_{2i}$

Until: $E < \varepsilon$ or iter

iii) $S_1 = 1/\tilde{S}_1$, $S_2 = 1/\tilde{S}_2$

Output: common reflectance R, temporal-1 illumination component S_1, temporal-2 illumination component S_2

III. Multitemporal Intrinsic Image Decomposition With Change Detection

Because there are always land cover changes between multitemporal remote sensing images and there is no common reflectance in the changing areas, the general framework of MIID proposed above cannot be directly used in the analysis of multitemporal remote sensing images in the changing areas. In order to make the proposed MIID method more applicable, we add CD steps to the general intrinsic decomposition framework to eliminate the interference of land cover changing. When extracting the reflectance of the changed areas, we must ensure that the reflectance of the changed areas in different temporal images has the same data distribution as the common reflectance of the unchanged area. Only in this way, the reflectance of the final formed unchanged areas and changed areas can meet the requirements of integrated processing. In order to solve these problems, we propose an MIID with CD (CD-MIID): firstly, we update the signal representation model of MIID and add a change mask. Secondly, a simple and effective multiple reflectance collaborative change vector analysis (CVA) method is proposed to achieve the accurate extraction of the changed areas. Thirdly, we build an improved intrinsic decomposition energy constraint model in the background consistency constraint of the changed areas,

$$\tilde{S}_{1i} \leftarrow \frac{\sum_{b=1}^{B} I_{1ib}R_{1ib}}{\sum_{b=1}^{B}(I_{1ib})^2} \tag{10}$$

$$\tilde{S}_{2i} \leftarrow \frac{\sum_{b=1}^{B} I_{2ib}R_{2ib}}{\sum_{b=1}^{B}(I_{2ib})^2} \tag{11}$$

$$R_{ib} \leftarrow \frac{2\left(\tilde{S}_{1i}\cdot(I_{1b})_i + \tilde{S}_{2i}\cdot(I_{2b})_i + 2R_{jb}\sum_{j\in N(i)} w_{ji}\right) - \sum_{j\in N(i)} w_{ji}\sum_{k\in N(j)} w_{jk}R_{kb}}{2\left(3 - R_{ib}\sum_{j\in N(i)} w_{ij}R_{jb}\right)} \quad \forall i \in P \ \forall b \in [1,\ldots,B] \tag{12}$$

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so that the reflectance extracted from the changed areas and the reflectance around the changed areas have the same data distribution.

A. CD-MIID Hypothesis

When the type of land cover is changed, the original multitemporal signal expression model (2) is no longer suitable. In order to adapt to the change of land cover, we added the change mask (M) to the original MIID signal model. The new signal expression model is as follows:

$$\begin{aligned} I_1 &= S_{1U} R_U M + S_{1C} R_{1C} (1 - M) \\ I_2 &= S_{2U} R_U M + S_{2C} R_{2C} (1 - M) \end{aligned} \tag{13}$$

where I_1 and I_2 represent the two input multitemporal images, R_U is the common reflectance of the unchanged areas of the two images (I_1 and I_2), R_{1C} is the reflectance of the changed areas of temporal-1 images I_1, R_{2C} is the reflectance of the changed areas of temporal-2 images I_2, S_{1U} and S_{2U} denote the illumination components of the unchanged areas of I_1 and I_2, respectively, and S_{1C} and S_{2C} denote the illumination components of the changed areas of I_1 and I_2, respectively.

B. Model Simplification and Change Mask Calculation

At present, many CD methods have been proposed or used to extract change mask between multiple remote sensing images, including classical principal component analysis, CVA [45], multivariable alternation detection (MAD) [46], and so on. In recent years, kernel regression method [47] and deep learning method are widely used in CD, especially the CD between different VHR images [23], [25], [48], between multisensor images [48] or cross-scene images [23].

Because it is difficult to invert too many unknown variables in a model at the same time, in order to reduce the inversion difficulty of the above model, we simplify the solution method of the above model (13), that is, take out R_U, S_{1U}, S_{2U} and M to solve separately. The three variables R_U, S_{1U}, S_{2U} can be obtained directly from the MIID method if M is known.

As we all know, the change of the surface material is mainly reflected in the spectral change, so the reflectance component of the intrinsic decomposition is easier to distinguish the changing area of the surface material. Based on this, we construct a CD method of multireflectance collaborative CVA. The core of this method is to analyze the difference of reflectance vectors of common reflectance and independent reflectance of each image at the same time. For any point i, the mask value of point i is calculated as follows:

$$\begin{aligned} M_i &= \mathrm{CVA}(R_i, R_{1i}) \cdot \mathrm{CVA}(R_i, R_{2i}) \\ &= \sqrt{\sum_{b=1}^{B} (R_{ib} - R_{1ib})^2 \sum_{b=1}^{B} (R_{ib} - R_{2ib})^2}. \end{aligned} \tag{14}$$

By setting minimum-cost threshold [49] on M_i, the mask of change areas can be easily obtained. After that, the other three variables (R_U, S_{1U}, and $S_{2\,U}$) can also be obtained through the MIID methods. At this point, the inversion of the variables in (13) finally comes down to the inversion of their respective changing areas.

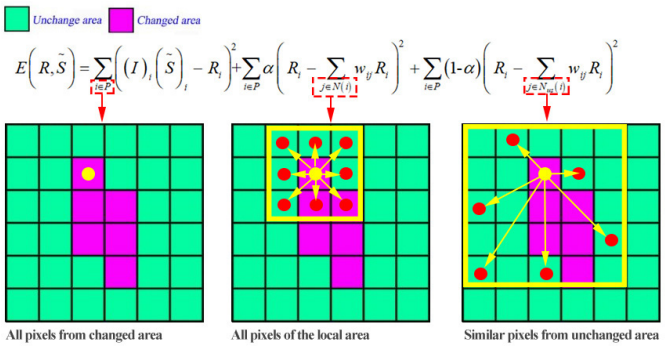


Fig. 2. Background consistency constraint for changed area.

C. Multitemporal Consistency Decomposition for Change Area

Through the above model simplification, the solution of the CD-MIID model finally falls on the intrinsic decomposition of change areas on each image. Before building the energy constraint model and iterative optimization model of the change areas, we firstly introduce the following three aspects of consensus:

The signal synthesis model still obeys the strand intrinsic decomposition model, so the product of the signal components should be close to the original signal, i.e., the error between the product of common reflectance and their own illumination and the original image should be as small as possible.

1) The reflectance of the pixel from the changing area should be close to the similar pixel from its local neighborhood. So, the reflectance of changing pixel can be expressed as the linear group of its local neighborhood pixels.
2) In order to ensure that the reflectance of the changed area is consistent at data distribution with the common reflectance of the unchanged area obtained by MIID method, the reflectance of the change pixel should be similar to that of the similarity pixels of the surrounding unchanged area, so the reflectance of the change pixel should also be expressed by the linear combination of the similarity point of the surrounding unchanged area.

In view of the above three consensuses, we build a background consistency energy constraint model for multitemporal changing area intrinsic decomposition. For the changed area reflectance of each image, the energy function is expressed as follows:

$$\begin{aligned} E(R, \tilde{S}) = &\sum_{i \in P} \left((I)_i \left(\tilde{S}\right)_i - R_i\right)^2 + \sum_{i \in P} \alpha \left(R_i - \sum_{j \in N(i)} w_{ij} R_i\right)^2 \\ &+ \sum_{i \in P} (1 - \alpha) \left(R_i - \sum_{j \in N_{sn}(i)} w_{ij} R_i\right)^2 \end{aligned} \tag{15}$$

where I is the original change area image of input image (temporal-1 image or temporal-2 image), and $N_{sn}(i)$ stands for the similar pixels in the unchanged area in local windows. α is used to balance the similarity constraint in the local neighborhood and background consistency energy constraint of the unchanged pixels. Generally, α is set to 0.5. Fig. 2 shows

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the relationship between energy function and each energy constraint term of (15). The first term of energy function is the relationship constraints of image composition [i.e., (1)]. In this item, only the points in the change area participate in the calculation. The second term of energy function is the similarity constraint of reflectance in a local neighborhood. For each point in the changing region, when calculating this constraint energy, its range is all points in the local neighborhood. It is exactly the same as the second term of (3). This part mainly guarantees the local similarity. The third term of energy function constrains the background consistency. For each point in the changing region, when calculating this constraint energy, its range is some similar pixels from nearby unchanged region. This part mainly ensures that the reflectance of the changed area is consistent at data distribution with the common reflectance of the unchanged area. The schematic of the energy model is as follows.

Similarly, the energy optimization of (15) is also a typical quadratic equation based on the variables R and $\tilde{S}$. By obtaining the first-order differentiation of (15), we can get

$$\frac{\partial E(R,\tilde{S})}{\partial \tilde{S}} = 2\sum_{b=1}^{B} I_b\left(\tilde{S}I_b - R_b\right) \tag{16}$$

$$\begin{aligned}\frac{\partial E(R,\tilde{S})}{\partial R_b} = {} & 4R_{ib} + 2\sum_{j\in N(i)} w_{ji}\sum_{j\in N(i)} w_{jb}R_{jb} \\ & + 2\sum_{j\in N_{us}(i)} w_{ji}\sum_{j\in N_{us}(i)} w_{jb}R_{jb} \\ & - 2\sum_{j\in N(i)} w_{ij}R_{jb} - 2\sum_{j\in N_{us}(i)} w_{ij}R_{jb} \\ & - 2\sum_{j\in N(i)} w_{ji}R_{jb} - 2\sum_{j\in N_{us}(i)} w_{ji}R_{jb} \\ & \forall i \in P \ \forall b \in [1,\ldots,B]\end{aligned} \tag{17}$$

$$\frac{\partial E(R,\tilde{S})}{\partial \tilde{S}} = \frac{\partial E(R,\tilde{S})}{\partial R_1} = \cdots = \frac{\partial E(R,\tilde{S})}{\partial R_b} = \cdots = \frac{\partial E(R,\tilde{S})}{\partial R_B}. \tag{18}$$

By finding the zero position of the first derivative of (16) and (17) [i.e., building (18)] and using the Gaussian–Seidel method, we can iteratively optimize (15) and get the optimal solutions of CD-MIID. The iteration functions are shown as follows:

$$\tilde{S}_i \leftarrow \frac{\sum_{b=1}^{B} I_{ib}R_{ib}}{\sum_{b=1}^{B}(I_{ib})^2} \tag{19}$$

$$\begin{aligned}R_{ib} \leftarrow 0.5\Bigg(& \sum_{j\in N(i)} w_{ij}R_{jb} + \sum_{j\in N_{us}(i)} w_{ij}R_{jb} \\ & + \sum_{j\in N(i)} w_{ji}R_{jb} + \sum_{j\in N_{us}(i)} w_{ji}R_{jb}\Bigg)\end{aligned}$$

$$\begin{aligned}& - 0.5\Bigg(\sum_{j\in N(i)} w_{ji}\sum_{j\in N(i)} w_{jb}R_{jb} \\ & + \sum_{j\in N_{us}(i)} w_{ji}\sum_{j\in N_{us}(i)} w_{jb}R_{jb}\Bigg).\end{aligned} \tag{20}$$

The detailed procedure of CD-MIID is shown as follows.

Algorithm 2 Procedure of CD-MIID

Input: Temporal-1 image I_1, Temporal-2 image I_2, $\varepsilon > 0$
i) Calculate R, S_1 *and* S_2 using MIID method
ii) Calculate change mask M using (14) and Ref [48]
iii) $R_U = R \cdot M$
iv) For each pixel i at change area, finding pixel i local neighborhood $N(i)$, finding pixel i similar pixel $N_{sn}(i)$ from unchanged area, calculate w_{ij} using (4).
v) Repeat
  Calculate R_{1Ci} by using (20)
  Calculate $\tilde{S}_{1Ci}$ by using (19)
  Calculate E by using (15) with R_{1Ci} and $\tilde{S}_{1Ci}$
 Until: $E < \varepsilon$ or iter
vi) $S_{1C} = 1/\tilde{S}_{1C}$
vii) Calculate R_{2C} and S_{2C} using step v and vi
viii) $S_1 = S_{1U} \cap S_{1C}$, $S_2 = S_{2U} \cap S_{2C}$,
Output: change area mask M, common reflectance R_U of unchanged area, reflectance R_{1C} of temporal-1 changed area, reflectance R_{2C} of temporal-2 changed area, temporal-1 illumination component S_1, temporal-2 illumination component S_2

In order to get faster convergence to the optimal solution, the initial values of parameters (R, S_1, S_2, R_{1C}, S_{1C}, R_{2C}, S_{2C}, and ε) can be set as follows: R, S_1, S_2, and ε are set as MIID; S_{1c} is the average of each band image of temporal-1 image's changed area (only the change area); S_{2c} is the average of each band image of temporal-2 image's changed area (only the change area); S_2 is the average of each band image of temporal-2 image; $R_{1C} = (I_{1C}/S_{1C})$; $R_{2C} = (I_{2C}/S_{2C})$; and similarity, in order to ensure calculation time is not too long, we set the iteration number as 100.

IV. Data and Experimental Design

A. Simulated Multitemporal Image With Reflectance Truth

To quantitatively evaluate and compare MIID approaches with other IID methods, the tested multitemporal remote sensing images should have a ground-truth reflectance image. However, it is different to obtain real reflectance components from remote sensing scenes, and until now, there is no real reflectance image of remote sensing image indeed. Therefore, in order to achieve quantitative evaluation and comparison of algorithms, we used some simulated multitemporal images here. These simulated multitemporal images are from the MIT intrinsic dataset [50]. In this dataset, the same object images with multiple lighting conditions can be treated as multitemporal images, and each object has a ground-truth reflectance image. The reflectance truth maps are from actual

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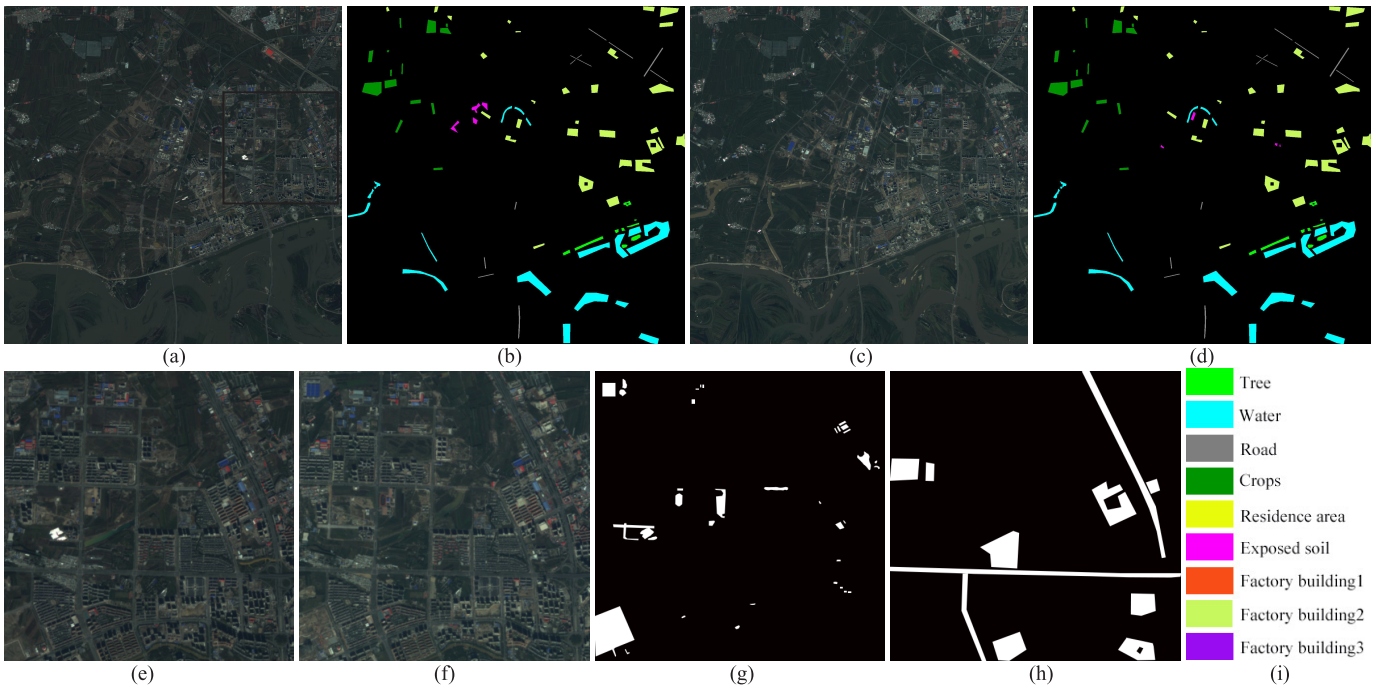


Fig. 3. Actual GF-1 multitemporal images. (a) Temporal-1 RGB image. (b) Ground truth of temporal-1. (c) Temporal-2 RGB image. (d) Ground truth of temporal-2. (e) and (f) Small region of the multitemporal image [red box in (a)]. (g) Changed area mask for the small region. (h) Unchanged area mask for small region. (i) Labels.

instruments measurement. In this article, we use two groups of simulated multitemporal images for testing: box and turtle. Relevant images can be downloaded easily from the website of MIT intrinsic dataset [51].

B. Actual Multitemporal Remote Sensing Images

The proposed MIID methods were implemented on four groups of actual multitemporal multispectral/hyperspectral remote sensing images: GF-1, GF-2, GF-5, and Landsat TM.

China put forward its High-Resolution Earth Observation System (CHEOS) in 2006. An important part of CHEOS is the high-resolution satellites. GF-1, GF-2, and GF-5 are belonging to high-resolution satellites in CHEOS. GF-1 and GF-2 are panchromatic/multispectral VHR satellites, mainly including four common visible/NIR bands (blue, green, red, and NIR), with resolutions of 2 m/8 m and 3.2 m/0.8 m, respectively. GF-5 is a hyperspectral imaging satellite, and it can collect 30-m resolution hyperspectral image with 330 spectral bands (150 in visible range, 180 in short wave range).

The first dataset collected from GF-1, which includes two multispectral images. The acquisition area is the suburb of Harbin city. The acquisition time is September 3, 2013 and September 20, 2014, and the image size is 1536 × 1536. The RGB images, ground truth, and changed and unchanged area mask (a small region, 512 × 512, located at the red tangle) of GF-1 dataset are shown in Fig. 3.

The second dataset collected from GF-2, which includes two multispectral images also. The acquisition area is part of Dalian city. The acquisition time are September 6, 2015 and June 28, 2016, and the image size is 2934 × 450. The RGB images, ground truth, and changed and unchanged area mask (a small region, 450 × 450, located at the red tangle) of the GF-2 dataset are shown in Fig. 4. The truth maps of GF-1 and GF-2 datasets are mainly obtained by manual annotation with the aid of auxiliary data (historical images of Google Earth).

GF-5 is the only satellite with hyperspectral imaging capability in CHEOS. When we used, some repetitive bands and edge bands are removed, so only 295 bands were used here. The GF-5 dataset (i.e., the third dataset) has two temporal images (731 × 760) collected over the Yellow River Wetland Park, Shandong province, China, on November 1, 2018 and May 31, 2019. The RGB images, missing data mask, ground truth (9 classes of ground cover), and changed area mask of the GF-5 dataset are shown in Fig. 5. The truth maps of GF-5 dataset come from field investigation (by The First Institute of Oceanography, MNR, China).

Landsat TM is one of the most widely used remote sensing data. It has seven spectral bands (30-m resolution) and one panchromatic band (15-m resolution). The Landsat TM dataset (i.e., the fourth dataset) used in this article has two temporal images (512 × 512) collected over Williston city, North Dakota state, USA, on July 28, 2006 and August 28, 2011. The RGB images, ground truth, and change area mask of the Landsat TM dataset are shown in Fig. 6. The truth maps of Landsat TM dataset are from the National Land Cover Database of the USA (https://www.mrlc.gov/).

C. Experimental Design

Before the MIID methods proposed in this article are used, their effectiveness and superiority should be tested. The experimental design is divided into two parts: testing MIID on simulated multitemporal images (with reflectance truth map) and testing MIID on actual multitemporal images.

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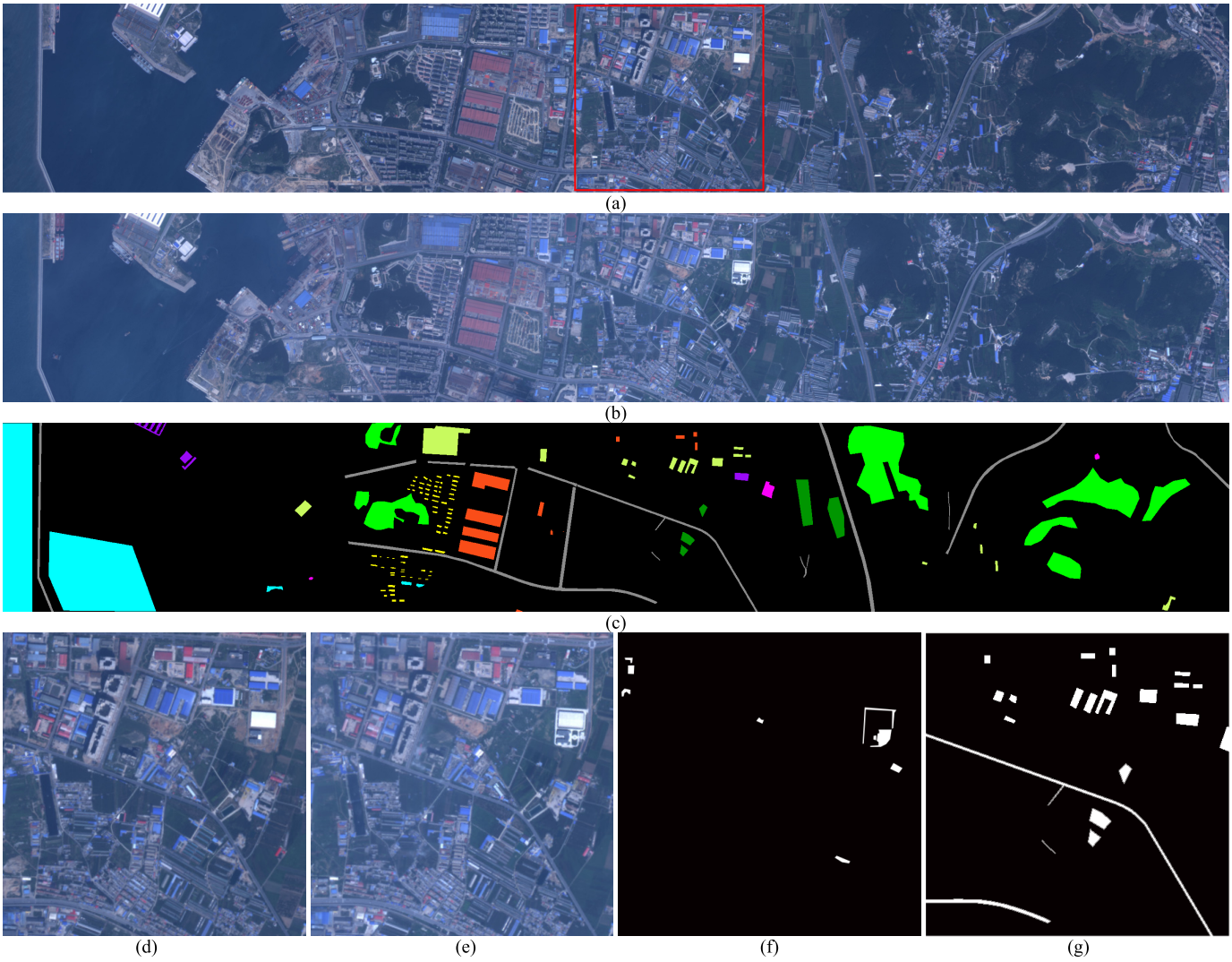


Fig. 4. Multitemporal GF-2 images. (a) Temporal-1 RGB image. (b) Temporal-2 RGB image. (c) Ground truth. (d) and (e) Small region of GF-2 multitemporal images [red box in (a)]. (f) Changed area mask for the small region. (g) Unchanged area mask for the small region.

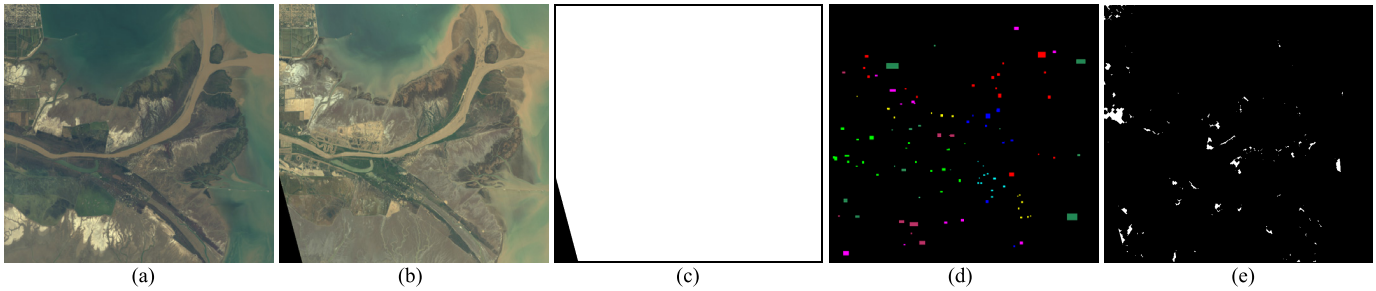


Fig. 5. Multitemporal GF-5 images. (a) Temporal-1 RGB image. (b) Temporal-2 RGB image. (c) Missing data mask. (d) Ground truth. (e) Changed area mask.

1) Experiment 1, Testing on Simulated Multitemporal Images: The proposed MIID methods were firstly tested on simulated multitemporal images (with reflectance truth map), which helps to understand its accuracy quantitatively at the aspect of reflectance. Three methods of intrinsic image decomposition using optimization (IID-OP) [32], intrinsic image decomposition by maximum likelihood estimation [38] (IID-ML, extracting reflectance from multiillumination synthetic image), and MIID were implemented for comparison purposes at the aspect of reflectance extraction. IID-OP, as a representative of the single image IID method, will be tested on each temporal image to ensure that the MIID have better performance than the IID on each individual image. Three objective performance measures are used for qualitative assessment: mean absolute distance (MABD), mean square error (MSE), and local MSE (LMSE) [52]. The MABD, MSE,

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(a) (b) (c) (d) (e)

Fig. 6. Multitemporal Landsat TM images. (a) Temporal-1 RGB image. (b) Ground truth of temporal-1. (c) Temporal-2 RGB image. (d) Ground truth of temporal-2. (e) Changed area mask.

and LMSE metrics are defined as follows:

$$\mathrm{MABD}\left(I_{\mathrm{GT}}, \widehat{I}\right) = \arg\min_{a} \left| I_{\mathrm{GT}} - a\widehat{I} \right| \tag{21}$$

$$\mathrm{MSE}\left(I_{\mathrm{GT}}, \widehat{I}\right) = \arg\min_{a} \left\| I_{\mathrm{GT}} - a\widehat{I} \right\| \tag{22}$$

$$\mathrm{LMSE}\left(I_{\mathrm{GT}}, \widehat{I}\right) = \sum_{w} \arg\min_{a} \left\| I_{\mathrm{GT}}^{w} - a\widehat{I}^{w} \right\|. \tag{23}$$

All the three indices mentioned above present degree of error between the reflectance extraction values and the actual reflectance values. Therefore, a smaller index value suggests a better extraction.

For the qualitative assessment of reflectance extraction, the inverted images were visually checked to determine whether they were closer to the reflectance truth maps, especially in terms of color consistency.

2) Experiment 2, Testing on Actual Multitemporal Images: The MIID methods were also applied to actual multitemporal remote sensing images to evaluate performance of reflectance component extraction and to test its application potential in multitemporal image analysis (include multitemporal CD and multitemporal classification). Three methods of IID-OP [32] (on each temporal image), IID-ML [38], and our MIID were implemented for comparison purposes at the aspect of reflectance extraction. For the qualitative assessment of reflectance extraction, the inverted images were visually checked to determine overall effect and color consistency with original images. Due to the lack of the reflectance truth map as the standard, it is impossible to evaluate the reflectance quantitatively. However, we can evaluate the application potential for the extracted reflectance through the accuracy of multitemporal classification. Seven traditional classification methods and multitemporal classification methods: 1) classify original target image of itself by KNN; 2) classify target reflectance image (extracted by IID-OP) with itself by KNN; 3) directly classify original target image with original source image by KNN; 4) classify original target image with original source image by TCA domain adaptation method; 5) directly classify target reflectance with source reflectance (extracted by IID-OP); 6) classify target reflectance with source reflectance by TCA domain adaptation method; and 7) classify reflectance extracted from MIID by KNN were implemented for comparison. Overall accuracy (OA) and classification map are used for qualitative and quantitative evaluation.

Seven CD methods of CD-PCA, CD-CVA [45], CD-MAD [46], CD-AMIR [47] (CD based on the comparison of affinity matrices and image regression), CD-IID-CVA (using reflectance images extracted by IID-OP method as the input of CVA method), CD-CVA-RRC (using images after relative radiometric correction [53] as the input of CVA method), and our proposed CD-MIID were implemented for comparison purposes at the aspect of change area extraction, and two objective performance measures are used here for qualitative assessment: receiver operating characteristic (ROC) curve and detection rate under fixed low false alarm rate. For the qualitative assessment of CD maps, the inverted images were visually checked to determine whether they were closer to the changed area truth maps.

V. Results and Analysis

A. Experiment Results of Simulated Multitemporal Images

Fig. 7(a)–(d) shows the reflectance images extracted by IID-OP with temporal-1 image, IID-OP with temporal-2 image, IID-ML, and our MIID method on the simulated multitemporal data-box. In Fig. 7(a) and (b), most of radial shadow and texture have been well removed by IID-OP, but some exceptions still exist. In Fig. 7(a), the flower color is changed from true crimson to light red, and there are some bright areas inside the mouth. In Fig. 7(b), the shadow in the top right corner of the box is not removed. For IID-ML, only part of the brightness is improved, but most of the shadows and texture still exist, so the effect of this method is the worst. In general, our method has more superior performance than other methods in box dataset.

Fig. 8(a)–(d) shows the reflectance images extracted by IID-OP with temporal-1 image, IID-OP with temporal-2 image, IID-ML, and our MIID method on the simulated multitemporal data-turtle. In Fig. 8(a), the brightness of turtle tail is seriously insufficient. In Fig. 8(b), a small amount of texture on the back of the turtle has not been eliminated. For IID-ML, the extracted reflectance image still similar to the original input image and the most of the shadows and texture still exists, so the effect of this method is still the worst. In general, our method has more superior performance than other methods on turtle dataset.

Table I shows the comparisons of reflectance extraction errors (MABD, MSE and LMSE) on the simulated multitemporal data. Obviously, our proposed MIID method always

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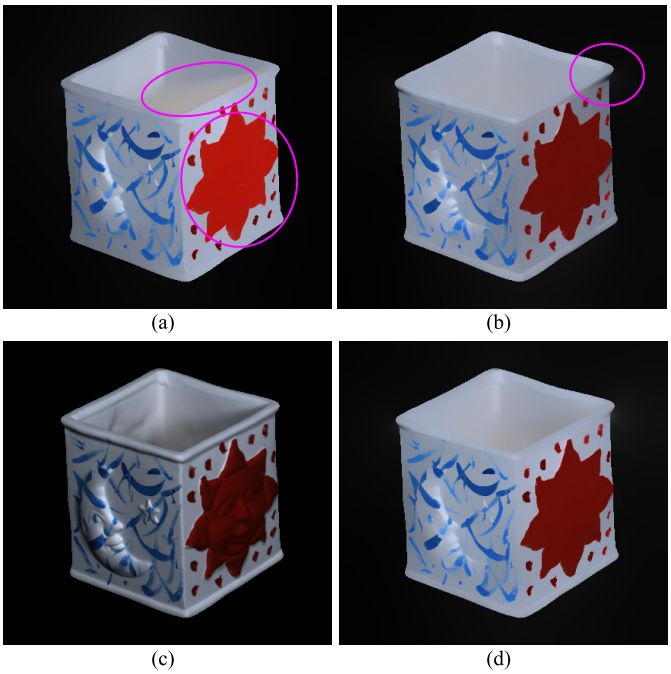


Fig. 7. Reflectance extraction results on simulated multitemporal data-box. (a) Temporal-1 reflectance by IID-OP method. (b) Temporal-2 reflectance by IID-OP method. (c) Common reflectance by IID-ML method. (d) Common reflectance by our MIID method.

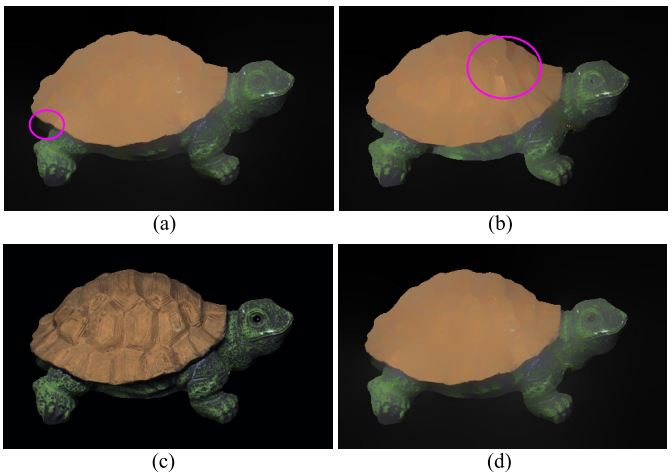


Fig. 8. Reflectance extraction results on simulated multitemporal data-turtle. (a) Temporal-1 reflectance by IID-OP method. (b) Temporal-2 reflectance by IID-OP method. (c) Common reflectance by IID-ML method. (d) Common reflectance by our MIID method.

has the highest extraction accuracy (i.e., smallest index value) in most cases compared with the other methods.

Algorithm Convergence Analysis: Fig. 9 shows the convergence speed of our MIID algorithm. The LMSE similarity index is used to evaluate the convergence. It can be seen that the proposed algorithm is convergent both on the two simulate data. The first 15 iterations converge faster, and after 40 iterations, the algorithm has basically converged to the optimal value.

To sum up, our MIID method has the best effect on the simulation multitemporal datasets which have real reflectance truth maps in terms of reflectance extraction.

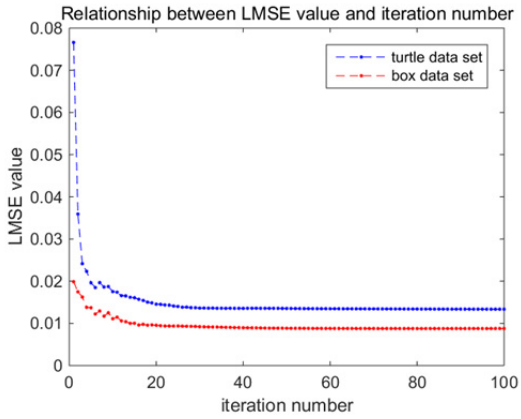


Fig. 9. Convergence speed of the MIID algorithm (relationship between the LMSE value and iteration number).

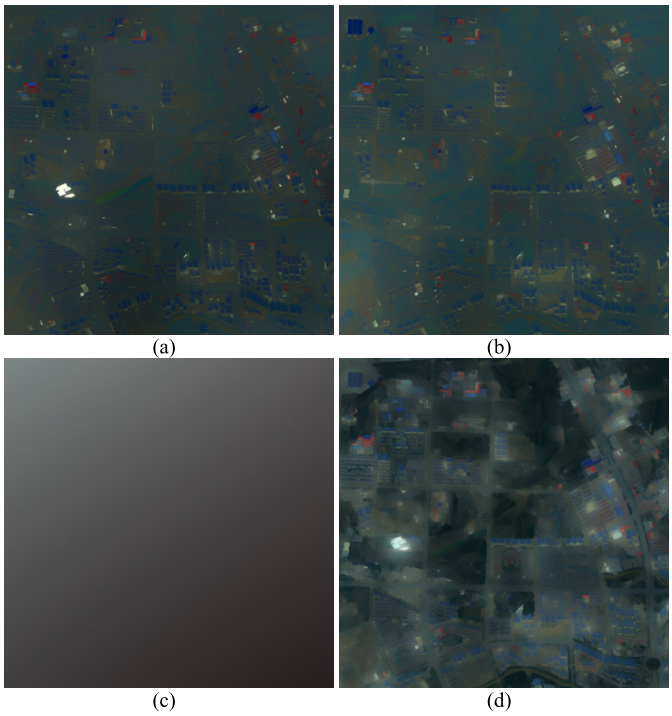


Fig. 10. Reflectance extraction results on actual multitemporal GF-1 images (at small region). (a) Temporal-1 reflectance by IID-OP. (b) Temporal-2 reflectance by IID-OP. (c) Common reflectance by IID-ML. (d) Common reflectance by MIID.

B. Experiment Results of Actual Multitemporal Images

Fig. 10(a)–(d) shows the reflectance images [only show the RGB bands and only use the small region shown in Fig. 3(a)] extracted by IID-OP with temporal-1 image, IID-OP with temporal-2 image, IID-ML, and our MIID method on the multitemporal GF-1 dataset. In Fig. 10(a) and (b), the color of the buildings highlighted in the image is the same as the original image, but the road cannot be found in both Fig. 10(a) and (b). Besides, abnormal light blue appears in many places, which is quite different from the original image color [Fig. 3(e) and (f)]. For IID-ML, reflectance color of land cover is totally missed, and the result cannot be used in practical application. For our method, the reflectance color of land cover is more similar than other methods and the

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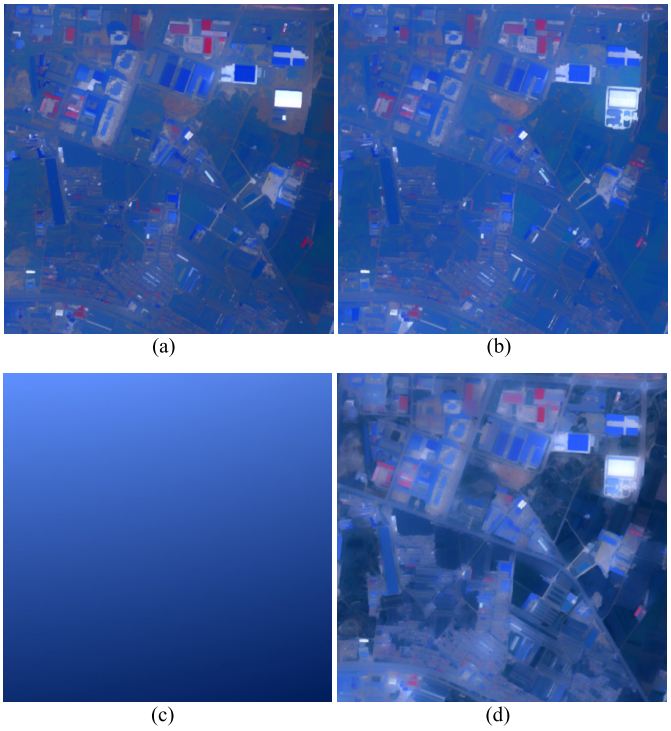


Fig. 11. Reflectance extraction results on actual multitemporal GF-2 images (at small region). (a) Temporal-1 reflectance by IID-OP. (b) Temporal-2 reflectance by IID-OP. (c) Common reflectance by IID-ML. (d) Common reflectance by MIID.

road area is clearly visible. Therefore, MIID has superior performance than other methods on GF-1 dataset.

Fig. 11(a)–(d) shows the reflectance images [only show the RGB bands and only use the small region shown in Fig. 4(a)] extracted multitemporal GF-2 dataset. All the results are similar to that of Fig. 10. Road cannot be found both in Fig. 11(a) and (b), and abnormal light blue appears in many places. Reflectance color of land cover is totally missed in Fig. 11(c). The color of the extracted reflectance by MIID method is closer to original images [Fig. 4(d) and (e)] than that extracted by other methods. Therefore, the performance of our MIID method on GF-2 dataset is still better than other methods.

Fig. 12(a)–(d) shows the reflectance images extracted from multitemporal GF-5 dataset. All the results are similar to that of Figs. 10 and 11. The color of the extracted reflectance by our MIID method is closer to original images [Fig. 5(a) and (b)] than that extracted by other methods. Therefore, the performance of our MIID method on GF-5 dataset is still better than other methods. Besides, it should be noticed that GF-5 is a hyperspectral satellite and our method is still effective and advanced in the hyperspectral dataset.

Fig. 13(a)–(d) shows the reflectance images extracted from multitemporal Landsat dataset. All the results are similar to that of Figs. 10–12. The color of the extracted reflectance by our MIID method is closer to original images [Fig. 6(a) and (c)] than that extracted by other methods. Therefore, the performance of our MIID method on Landsat dataset is still better than other methods.

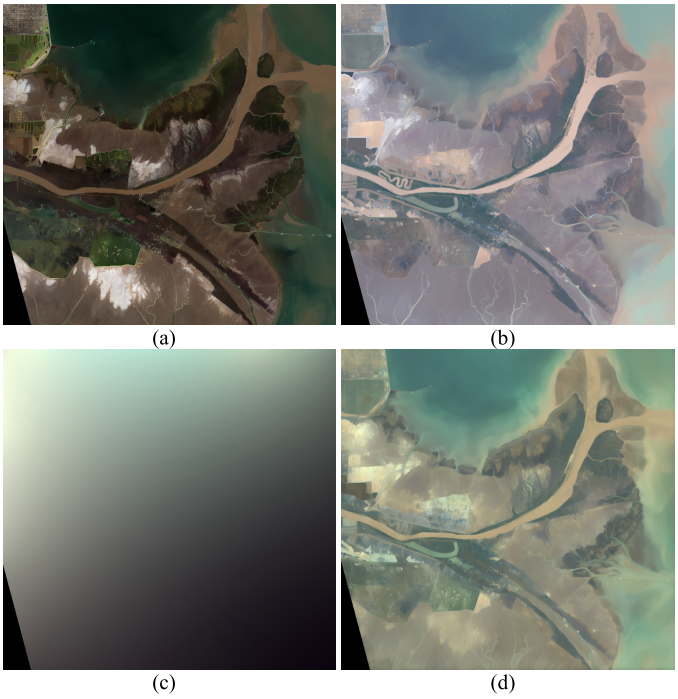


Fig. 12. Reflectance extraction results on actual GF-5 multitemporal images. (a) Temporal-1 reflectance by IID-OP. (b) Temporal-2 reflectance by IID-OP. (c) Common reflectance by IID-ML. (d) Common reflectance by MIID.

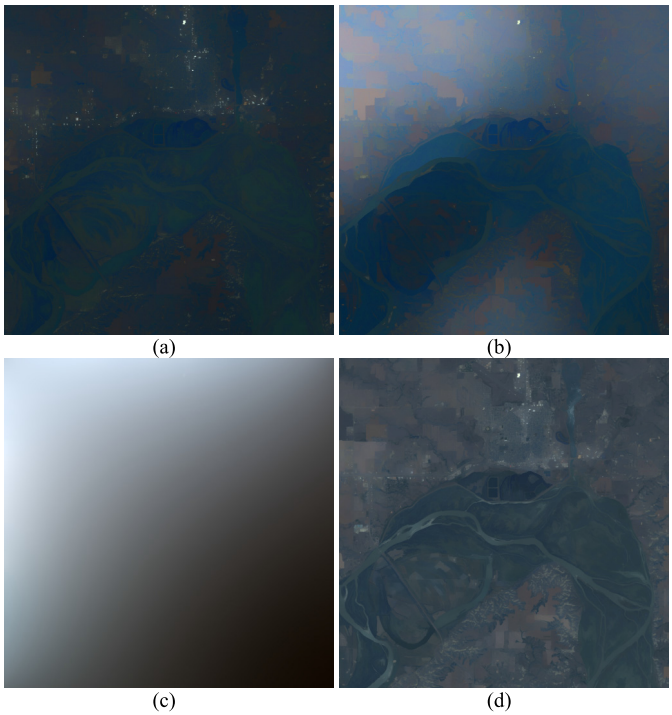


Fig. 13. Reflectance extraction results on actual multitemporal Landsat TM images. (a) Temporal-1 reflectance by IID-OP. (b) Temporal-2 reflectance by IID-OP. (c) Common reflectance by IID-ML. (d) Common reflectance by MIID.

In order to demonstrate our proposed MIID method is useful for multitemporal classification, we try to compare the direct classification and DA classification (using typical TCA method) on multitemporal original images and spectral

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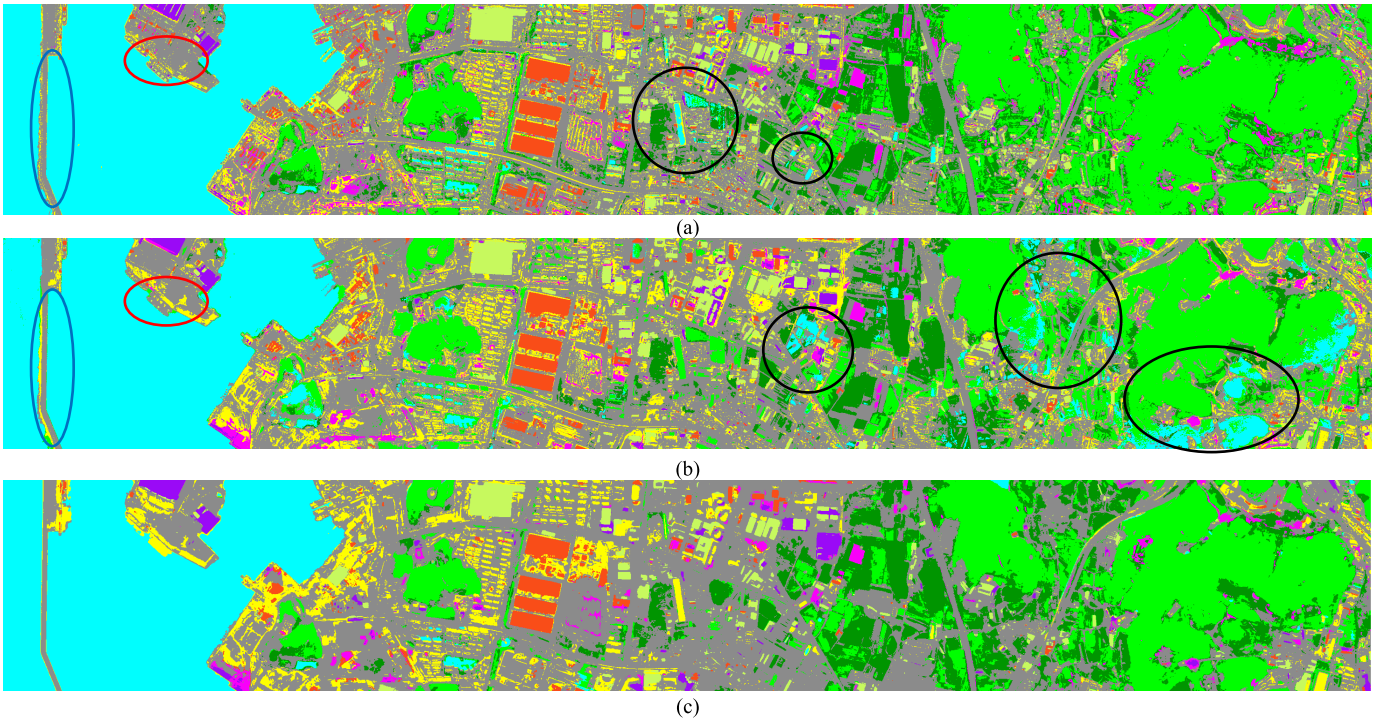


Fig. 14. Classification maps of GF-2 dataset. (a) Classification maps of original target image. (b) Classification maps of target image's reflectance extracted by IID-OP method. (c) Classification maps of common reflectance by our CD-MIID method.

TABLE I

ACCURACY OF REFLECTANCE EXTRACTION ON TWO GROUPS OF SIMULATED MULTITEMPORAL IMAGES

Measures Index	Data set	IID-OP1	IID-OP2	IID-ML	MIID
MABD	Box	0.2487	0.2212	0.3648	**0.1948**
	Turtle	0.1107	0.0785	0.0754	**0.0737**
MSE	Box	0.0806	0.0674	0.1684	**0.0523**
	Turtle	0.0161	**0.0082**	0.0092	0.0090
LMSE	Box	0.0100	0.0100	0.0130	**0.0088**
	Turtle	0.0139	0.0139	0.0175	**0.0134**

TABLE II

MULTITEMPORAL CLASSIFICATION ACCURACY (OA) WITH DIFFERENT TRAINING SAMPLE NUMBERS ON FOUR GROUPS OF ACTUAL MULTITEMPORAL DATASETS (%)

Data set	Sample number	Method-1	Method-2	Method-3	Method-4	Method-5	Method-6	Method-7
GF-1	50	81.29	79.40	31.30	28.04	29.75	20.12	**89.63**
	100	84.24	84.29	31.36	25.44	28.87	21.98	**92.20**
	200	86.38	87.87	31.33	23.90	25.56	20.06	**93.84**
GF-2	50	93.21	93.22	80.05	87.72	56.66	39.89	**94.39**
	100	94.44	95.06	81.59	89.37	59.46	42.00	**95.36**
	200	95.45	96.15	81.49	90.01	62.87	41.88	**96.75**
Landsat TM	1%	44.87	58.95	17.35	21.86	9.57	7.94	**61.84**
	2%	45.34	59.55	17.75	24.18	6.61	5.27	**62.98**
	3%	45.43	60.36	18.93	25.11	3.83	6.52	**63.43**
GF-5	50	84.50	89.12	25.44	35.14	28.92	36.53	**94.17**
	100	90.62	94.74	25.81	36.15	29.27	37.55	**98.81**
	200	94.10	98.20	26.44	34.35	30.44	37.90	**99.52**

PS: Method-1, classify original target image with itself by KNN; method-2, classify target reflectance image (extracted by IID-OP) with itself by KNN; method-3, directly classify original target image with original source image by KNN; method-4, classify original target image with original source image by TCA domain adaptation method; method-5, directly classify target reflectance with source reflectance (extracted by IID-OP); method-6, classify target reflectance with source reflectance by TCA domain adaptation method; method-7, classify reflectance extracted from MIID by KNN.

reflectance extracted images (totally seven methods, see the back of Table II or experiment setting above). The results (Table II) show that our MIID method has the highest classification accuracy in all cases. Here, we can get two conclusions: first, the performance of the proposed framework is much better than the traditional multitemporal classification, and also better than the classification accuracy of the target image self-classification; second, classification results of reflectance extracted by the IID method are better than the original image, and the classification accuracy of MIID is higher than IID. So, it proves that MIID can extract more accurate reflectance than IID.

Fig. 14(a)–(c) shows the classification maps of GF-2 dataset. Here, we only show the results of two comparison methods with the highest classification accuracy (target image self-classification and target reflectance image self-classification). Through analysis, it is found that many of these images [Fig. 14(a) and (b)] are mistakenly classified as water bodies (marked as black). Moreover, the classification results of the bridge on the left side are not as clean as our proposed method, and many parts of the bridge are wrongly classified as buildings (marked in blue). In addition, the results of our method are also very regular in some ports (marked in red). Therefore, our method has the best performance at the classification map aspect.

Fig. 15(a)–(c) shows the classification maps of GF-5 dataset. Some rivers (at the bottom, mark in black) which are hard to find in other methods can be clearly found in our classification map. Therefore, our method has the best performance on this hyperspectral dataset.

Fig. 16(a)–(g) shows the CD maps by CD-PCA, CD-CVA, CD-MAD, CD-AMIR, CD-IID-CVA, CD-CVA-RRC, and our proposed CD-MIID method on the multitemporal Landsat

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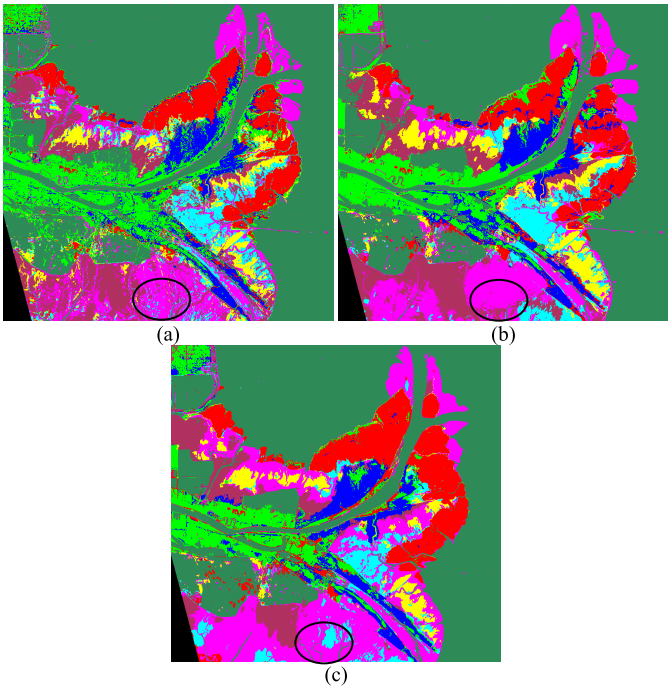


Fig. 15. Classification maps of GF-5 Hyperspectral data. (a) Classification maps of original target image. (b) Classification maps of target image's reflectance extracted by IID-OP method. (c) Classification maps of reflectance by our CD-MIID method.

TM dataset. In comparison with the truth map of change [in Fig. 6(e)], only our CD-MIID method can find most of the change areas at the Missouri River channel. CD-IID-CVA method also has a good CD effect, but the overall detection contrast is not obvious, which is easy to cause false alarm. CD-CVA-RRC method has a good CD effect in the main part of the river, but it does not detect the vertical river change.

Fig. 17(a)–(d) shows the CD maps by CD-PCA, CD-CVA, CD-MAD, CD-AMIR, CD-IID-CVA, CD-CVA-RRC, and CD-MIID methods on the multitemporal GF-5 dataset. In Fig. 17(a)–(c), the detection probability values of many places (marked in violet) which are obviously changed in the original images [Fig. 5(a) and (b)] are almost undetected, so there are many false detection areas in the comparison methods. Only our CD-MIID method detected those changes. Besides, the detection result of CD-PCA method looks reversed, most of the detection result of CD-MAD method is wrong, and the brightness of CD-AMIRs result is generally too high, resulting in high false alarm in later detection. Although CD-CVA-RRC method can detect the changes mentioned above, it also detects some unchanged areas (marked in violet). Therefore, our method has more superior performance than other methods on GF-5 data.

Table III shows the comparisons of CD probability at fixed low false positive rate (1.0×10^{-1}) on actual multitemporal datasets. It is easy to see that our CD-MIID method has the highest detection probability in Landsat TM and GF-5 cases compared with the other methods. Although the detection accuracy of our CD-MIID method is lower than that of CD-MAD method on GF-1 data and lower than that of CD-CVA-RRC method on GF-2 data, the detection accuracy

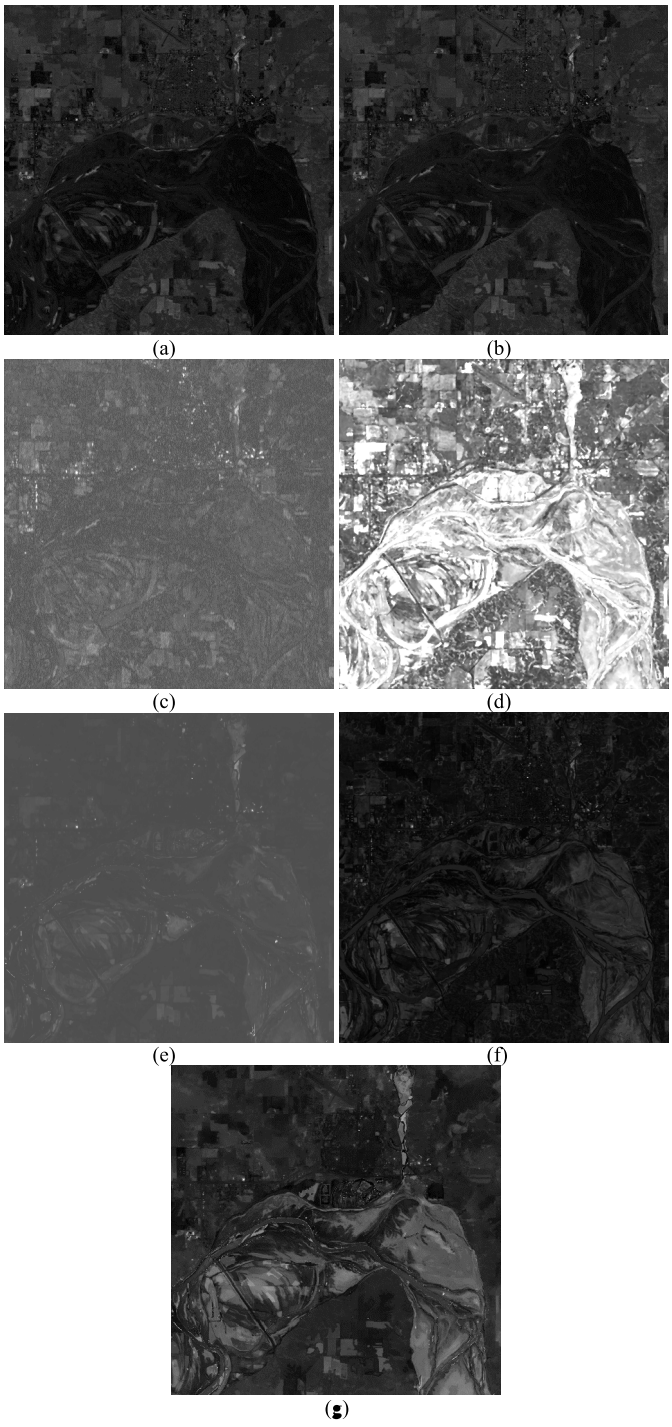


Fig. 16. CD results on actual multitemporal Landsat TM images. (a) CD-PCA method. (b) CD-CVA method. (c) CD-MAD method. (d) CD-AMIR method. (e) CD-IID-CVA method. (f) CD-CVA-RRC method. (g) our CD-MIID method.

is close. The reason why MIID is weaker than CD-MAD in GF-1 case is that the image acquisition date is basically the same, and the image difference is small. The traditional method works well at this time. In particular, the detection accuracy of our method on Landsat data is more than 40% higher than other methods. Therefore, our method is more effective on CD. It should be noted that the CD rate of GF-5 and Landsat dataset is generally low. This is mainly due to

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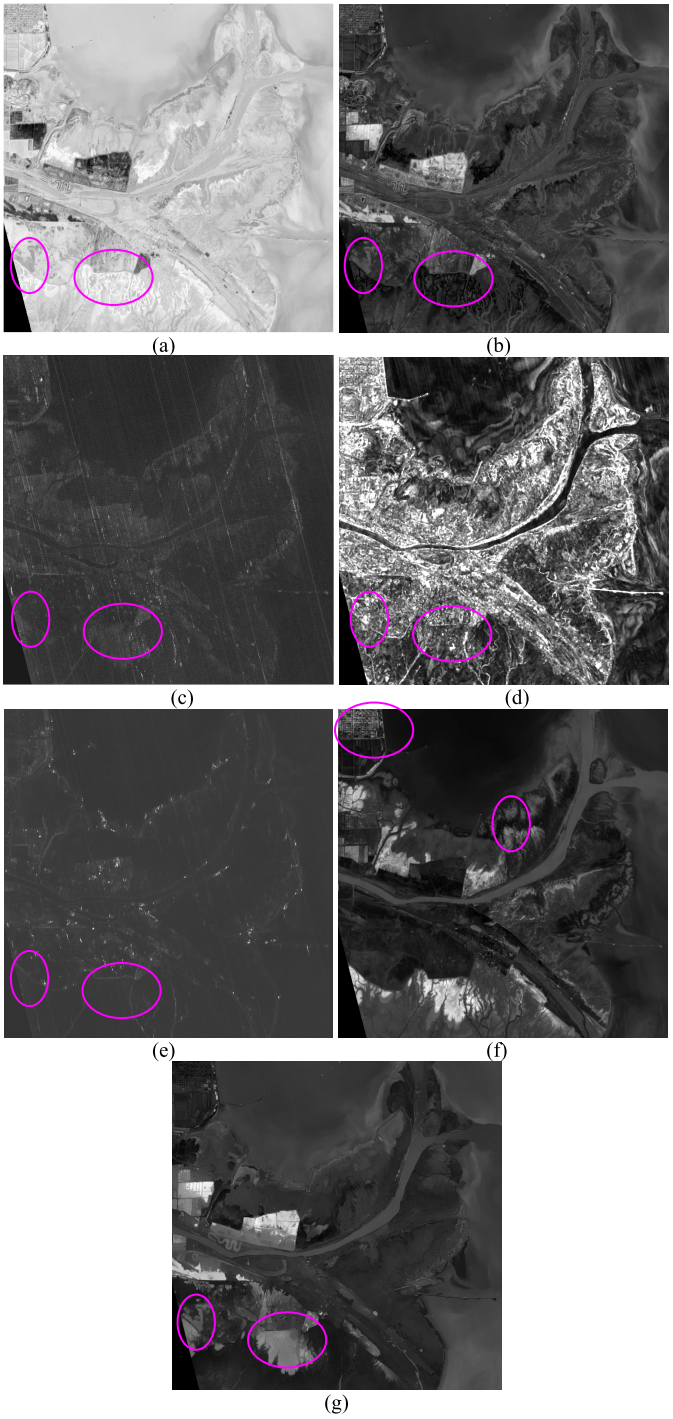


Fig. 17. CD results on GF-5 images. (a) CD-PCA method. (b) CD-CVA method. (c) CD-MAD method. (d) CD-AMIR method. (e) CD-IID-CVA method. (f) CD-CVA-RRC method. (g) Our CD-MIID method.

the low spatial resolution (30 m), and the changed pixels are often mixed pixels, so it is difficult to detect them.

Fig. 18 shows the ROC curves on the actual multitemporal datasets. It is also easy to see that our CD-MIID method has best detection probability in Landsat TM and GF-5 cases [Fig. 18(c) and (d)]. On GF-1 dataset [in Fig. 18(a)], our proposed CD-MIID method has the second-best performance, and will have the best performance as well as the CD-MAD method when the false positive rate is between 0.018 and

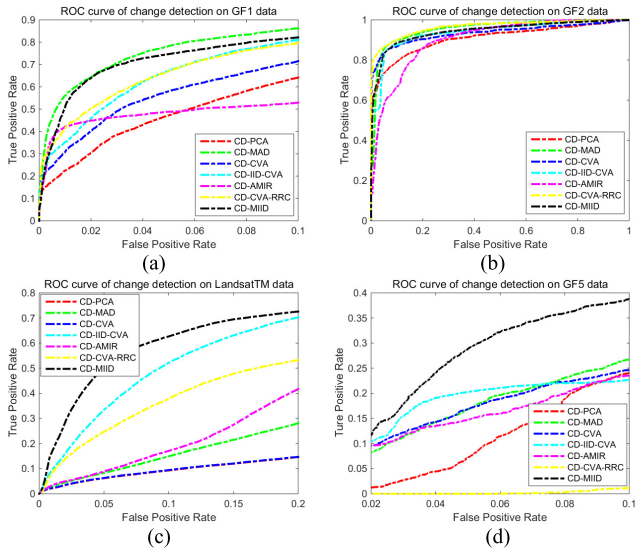


Fig. 18. ROC curves of CD on actual multitemporal datasets. (a) GF-1. (b) GF-2. (c) Landsat TM. (d) GF-5.

TABLE III

CD PROBABILITY ON FOUR GROUPS OF ACTUAL MULTITEMPORAL IMAGES AT 1.0×10^{-1} FALSE ALARM LEVEL

Data set	CD-PCA	CD-CVA	CD-MAD	CD-AMIR	CD-IID -CVA	CD-CVA -RRC	CD-MIID
GF-1	64.6%	71.9%	**87.1%**	53.5%	81.4%	79.8%	82.3%
GF-2	79.0%	86.9%	90.1%	69.1%	87.1%	**90.8%**	89.2%
Landsat	9.4%	9.4%	15.6%	17.7%	50.5%	37.4%	**63.4%**
GF-5	24.7%	24.9%	27.3%	23.9%	22.6%	1.4%	**38.8%**

TABLE IV

COMPUTATIONAL TIME OF EACH METHOD(S)

Data set	IID-OP1	IID-OP2	IID-ML	MIID
Box	185.94	188.14	0.96	272.02
Turtle	150.11	151.65	0.47	222.16
Landsat	388.73	423.14	1.83	504.98
GF-5	1975.96	2064.23	4.06	3182.14

0.025. On GF-2 dataset [in Fig. 18(b)], CD-CVA-RRC method has the best performance and our CD-MIID method has the third-best performance. However, it should be noticed that many methods have good performance on GF-2 data (including our method), and the ROC curves of these methods are very close, so the detection effect of our method is still acceptable.

To sum up, our CD-MIID method has a better performance at CD on actual multitemporal datasets, including multispectral and hyperspectral images.

C. Computing Time Comparison

Table IV shows the computing time of each reflectance extraction method on simulate data and actual multitemporal data. Here, time is measured in seconds on a laptop with 2.6G i7-6700HQ CPU and 8 GB RAM. It is easy to see that the main limitation of the proposed MIID method lie in its high computational cost as IID-OP method. The reason is that both IID-OP method and our MIID method require many iterations

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to find the optimal solutions (the number of iterations is set to 100). Besides, because our MIID method needs to optimize two temporal images at the same time, the computing time is higher than IID-OP method, but less than twice. Although computing time is higher than others, it is worth noting that our method can recover with more accurate reflectance results than others.

VI. Conclusion

This article focuses on how to extract more accurate spectral reflectance information from multitemporal remote sensing images and improves the existing multitemporal analysis system which is seriously affected by spectral drift. For this purpose, we have established a complete framework named MIID. We also improved MIID methods to make it suitable for the land cover change situation. After that, we also extended MIID methods so that it can be applied in the current multitemporal analysis system. The core of the MIID method is to add temporal–spatial energy constraints in traditional IID framework and construct the corresponding iterative optimization model. The core of the CD-MIID method is to extract change mask and change area's reflectance with the background consistency constraint. The results of the experiment confirm that our methods are better than the traditional method in the accuracy of reflectance extraction, multitemporal classification, and CD. However, like many other articles, our approach is also hardly to deal with the shadow problem between VHR images and the summer–winter case. Besides, some other shortcomings will be improved in our future developments, including: 1) it cannot deal with more temporal image case; 2) in some hyperspectral data, the ability of CD needs to be improved; and 3) the algorithm is time-consuming, and the speed needs to be improved.

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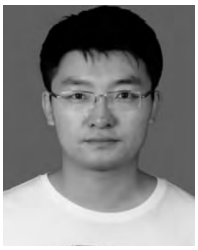
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